5.21. Schur polynomials

Let $\lambda = (\lambda_1, \ldots, \lambda_p)$ be a partition of n, and let $N \geq p$. Let

$$D_\lambda(x) = \sum_{s \in S_N} (-1)^s \prod_{j=1}^{N} x_{s(j)}^{\lambda_j + N - j} = \det(x_i^{\lambda_j + N - j}).$$

Define the polynomials

$$S_\lambda(x) := \frac{D_\lambda(x)}{D_0(x)}$$

(clearly $D_0(x)$ is just $\Delta(x)$). It is easy to see that these are indeed polynomials, as D_λ is antisymmetric and therefore must be divisible by Δ. The polynomials S_λ are called the **Schur polynomials**.

Proposition 5.21.1.

$$\prod_m (x_1^m + \cdots + x_N^m)^{i_m} = \sum_{\lambda : p \leq N} \chi_\lambda(C_{\mathbf{i}}) S_\lambda(x).$$

Proof. We will prove that

$$\Delta(x) \prod_{m \geq 1} (x_1^m + \cdots + x_N^m)^{i_m} = \sum_{\lambda : p \leq N} \chi_\lambda(C_{\mathbf{i}}) D_\lambda(x)$$

(where the sum is over partitions of n with at most N parts) by comparing the coefficients of x^ν for all $\nu = (\nu_1, \ldots, \nu_N)$. Both sides are antisymmetric polynomials, so it is enough to consider the case in which $\nu_1 > \cdots > \nu_N$. Also, both sides are homogeneous polynomials of degree $(N-1) + (N-2) + \cdots + 1 + n$, so we may assume that $\nu_1 + \cdots + \nu_N$ equals this. Let $\rho = (N-1, \ldots, 1, 0)$; then $\nu = \mu + \rho$ where $\mu_1 \geq \cdots \geq \mu_N \geq 0$ and $\mu_1 + \cdots + \mu_N = n$.

The coefficient of x^ν on the left side is $\chi_\mu(C_{\mathbf{i}})$ by the Frobenius character formula. The only monomial in $D_\lambda(x)$ with strictly decreasing exponents is $x^{\lambda+\rho}$ (the $\sigma = 1$ term in the determinant), so the coefficient of x^ν in the right side is the coefficient of $x^{\mu+\rho}$ in $\sum_{\lambda : p \leq N} \chi_\lambda(C_{\mathbf{i}}) x^{\lambda+\rho}$, which again is $\chi_\mu(C_{\mathbf{i}})$: only the $\lambda = \mu$ term contributes. $\square$

Certain special values of Schur polynomials are of importance. Namely, we have